Rule-Based Monetary Policy in China

A Counterfactual Analysis Using the Caravello et al. (2025) Framework

Liming Lin

April 2025 — *Work in Progress*

Sciences Po Paris

Supervisor: Paul Bouscasse

Jury: Paul Bouscasse, Axelle Ferrière

Menu

Motivation and Question

Motivation

The puzzle. Since 2023, Chinese CPI inflation has hovered near zero, persistently undershooting the official targets set in the annual Government Work Report (GWR).

- GWR CPI targets 2023–2025: **3%, 3%, 2%**
- Realized CPI YoY 2023–2025: roughly **0%**
- Xi's speech at the 2024 Central Economic Work Conference first introduced the phrase "a reasonable rebound in the price level" (物价合理回升); the 2025 CEWC explicitly tied this objective to the conduct of monetary policy.

The question. What would have happened to the Chinese economy if the PBoC had strictly followed some form of rule-based policy during this period?

Why it matters.

- Quantifies the policy gap: how far was actual policy from target-consistent policy?
- Tests whether conventional interest rate policy is even capable of closing the inflation gap, given China's transmission mechanism.

Methodology

Methodology Roadmap

Following Caravello, McKay & Wolf (2025), the empirics-only counterfactual requires **four building blocks**:

(1) Narrative shocks
Chen et al. (2025)
Miranda-Agrippino et al. (2026) policy rule residuals

(2) HFI shocks
Daily FR007 surprises on policy event dates

(3) Baseline Projections
Reduced Form VAR estimated with 7-variable and Minnesota-prior

(4) Counterfactual
Loss-function optimization over the 2D shock space

Methodology Roadmap Explanation

- Blocks (1) and (2) provide **two distinct empirical IRFs** of the Chinese economy to monetary policy shocks — the policy "treatments" whose linear combinations span the feasible policy space.
- Block (3) provides the **baseline projection** of where the economy goes absent any policy intervention.
- Block (4) solves for the shock sequence that moves the baseline toward alternative paths, subject to a multi-objective loss function.
- Using Caravello et al. 2025's framework, instead of traditional counterfactuals based on structural models like DSGE, allows us to avoid the strong assumptions about the underlying economic structure, especially given the unique features of the Chinese institutional and economic environment.

Narrative Shocks: Estimated Policy Rule

Specification. A monthly policy rule for the 7-day interbank repo rate (FR007), combining elements from Chen, Xiao & Zha (2025) and Miranda-Agrippino, Nenova & Rey (2026):

$$\begin{aligned}\mathsf{FR007}_t \;=\; \alpha \;+\; \rho\cdot\mathsf{FR007}_{t-1} \;+\; \beta_\pi\cdot\mathsf{cpi\_gap}_{t-1} \\ \;+\; \beta_y^{+}\cdot\mathsf{gap\_pos}_{t-1} \;+\; \beta_y^{-}\cdot\mathsf{gap\_neg}_{t-1} \;+\; \beta_e\cdot\mathsf{fx\_gap}_{t-1} \;+\; \varepsilon_t^{\mathsf{MP}}\end{aligned}$$

Variable definitions.

- $\mathsf{cpi\_gap}_t = \pi_t - \pi_t^*$, where π_t^* is the annual CPI target from the Government Work Report
- $\mathsf{gdp\_gap}_t = g_t - g_t^*$, where g_t is a proxy to monthly real GDP YoY (see appendix: ▸ Stock–Watson construction) and g_t^* is the GWR growth target
- $\mathsf{gap\_pos}_t$, $\mathsf{gap\_neg}_t$: asymmetric split of the GDP gap around a 1pp threshold (see appendix: ▸ regime-switch motivation)
- $\mathsf{fx\_gap}_t$: post-2006 indicator (when fixed exchange rate regime ended) interaction between the change in the central parity rate and the spot–parity deviation

Narrative Shocks: Identification and Rationale

Identification. The residual $\widehat{\varepsilon}_t^{\mathrm{MP}}$ isolates the unsystematic component of FR007 movements — the narrative monetary policy shock series. ▸ actual vs. predicted FR007

Why this hybrid specification?

- CXZ2025 kept FR007 as the dependent variable but excluded the exchange rate; MANR2026 added the foreign exchange term but used the a Composite Monetary Policy Indicator on the left-hand side. ▸ CMPI vs. FR007
- We retain FR007 (the actual market rate that anchors HFI surprises downstream) while adopting MANR2026's FX specification and the lag terms of those gap variables, which mechanically make the policy rates forward-looking.

The Impulse Responses to the shocks (also the later HFI shocks and baseline projections) are estimated using a Bayesian SVAR approach with a Minnesota-type prior following Giannone, Lenza & Primiceri (2015), which was adopted by MANR2026 as well: ▸ Narrative Shocks IRFs

High-Frequency Shocks: FR007 Surprises on Policy Event Dates

Identification strategy. Following Das & Song (2022), we identify monetary policy surprises as the daily close-to-close change in the FR007 rate around dates on which the PBoC announces a change to one of its core policy instruments:

$$\text{shock}_t = \left(\text{FR007}_t - \text{FR007}_{t-1}\right) \cdot \mathbb{1}\{t \in \mathcal{E}\}$$

where $\mathcal{E}$ is the set of policy event dates.

Core policy instruments tracked. Event dates are collected from PBoC by Wind:

- 7-day reverse repo rate (OMO) — the primary short-term policy rate post-2016
- 1-year Medium-term Lending Facility (MLF) — introduced 2014, principal medium-term policy tool
- 1-year Loan Prime Rate (LPR) — replaced benchmark lending rate as the loan pricing anchor in August 2019
- Reserve Requirement Ratio for large financial institutions (RRR)
- Benchmark deposit and lending rates (pre-2015 era, before the rate liberalization)

High-Frequency Shocks: Identification and Rationale

Rationale. Only dates on which the PBoC makes changes to either of the policy instruments are considered are recorded. The reason is that the PBoC makes announcements for certain tools like OMO on a regular basis, which cluster in the later years in the sample and do not necessarily reflect a change in the policy stance.

Limitations. Das & Song actually used the 1-year Interest Rate Swap (IRS) based on FR007 instead of the FR007 itself to better represent the reactions from the financial markets. However, the access to this data and the more detailed one in hourly frequency have not been granted by the authority yet.
In addition, the PBoC's announcements also occur on the same day with other events, including the release of macroeconomic data or the announcements by the fiscal authority, which we also want to come back to revisit if data is available.

▸ monthly HFI shock series

▸ HFI IRFs

▸ IRF comparison: any vs. rate-change only

Reduced-Form Long-Term Projection: Bayesian VAR

Specification. A 7-variable monthly BVAR with $p = 6$ lags.

$$y_t = c + \sum_{\ell=1}^{6} A_\ell \, y_{t-\ell} + \sum_{k=1}^{4} d_k \, D_t^{\text{covid},k} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma_u)$$

where $D_t^{\text{covid},k}$ are monthly dummies for Jan–Apr 2020.

Variables in y_t (BVAR ordering, monthly).

$$y_t = \left(g_t,\, \mathrm{IP}_t,\, \pi_t,\, i_t,\, m_t,\, e_t,\, y_t^{US}\right)'$$

g_t: real GDP YoY; IP_t: Industrial Value Added YoY; π_t: CPI YoY; i_t: FR007; m_t: M2 YoY growth; e_t: RMB Real Effective Exchange Rate YoY; y_t^{US}: US industrial production YoY;
Variables reflecting the labor market condition were dropped due to quality and availability issues;
Other potential variables like consumption, net exports, and fixed asset investments were excluded because they are already captured by the construction of the monthly GDP proxy.

Reduced-Form Long-Term Projection

Estimation samples.

- Monthly estimation with COVID dummies included in all runs.
- A_ℓ estimated for sample period from 2002 to 2022, and 2025.

Baseline projection.

$$\hat{y}_{t^*+h} = \hat{c} + \sum_{\ell=1}^{6} \hat{A}_\ell \, \hat{y}_{t^*+h-\ell}, \qquad h = 1, 2, \ldots, H = 120$$

▸ forecast vs. actual (proxy) monthly GDP

Proposition 1 from Caravello et al.2025 establishes the counterfactual's validity using the Wold representation of the data. In practice, we follow their approach to compute BVAR forecasts, which is equivalent to the Wold representation under invertibility.

Solving the Counterfactual

Inputs. We take three objects from earlier blocks: baseline paths $\pi^{\text{base}}, y^{\text{base}}, i^{\text{base}}$ over a T-month window; transmission maps M_π, M_y, M_i linking narrative and HFI shocks to each variable; and policy targets π^* and y^*.

We solve for $\tilde{\nu} \in \mathbb{R}^{2T}$, containing one narrative shock and one HFI shock each month. Counterfactual paths are baseline plus transmission:

$$\tilde{\pi} = \pi^{\text{base}} + M_\pi \tilde{\nu}, \quad \tilde{y} = y^{\text{base}} + M_y \tilde{\nu}, \quad \tilde{i} = i^{\text{base}} + M_i \tilde{\nu}.$$

Loss Function The shock sequence is chosen to minimize some combinations of the following objectives:

$$\mathcal{L}(\tilde{\nu}) = \lambda_\pi \|\tilde{\pi} - \pi^*\|^2 + \lambda_y \|\tilde{y} - y^*\|^2 + \lambda_i \|\tilde{i} - \tilde{i}_{t-1}\|^2 + \lambda_e \|\tilde{e}\|^2.$$

Terms. The four terms penalize CPI-target miss, GDP-target miss, non-smooth or abrupt FR007 moves, and exchange-rate instability; the weights $\lambda_\pi, \lambda_y, \lambda_i, \lambda_e$ define the targeting rule; while the discount factor is set to 1 for simplicity following Caravello et al.2025.

Solving the Counterfactual

Solution. Because paths are linear in $\tilde{\nu}$ and the loss is quadratic, this is a stacked OLS problem solved by one linear-system :

$$\tilde{\nu}^* = \arg\min_{\tilde{\nu}} \mathcal{L}(\tilde{\nu}) = A_{stack} \backslash b_{stack}.$$

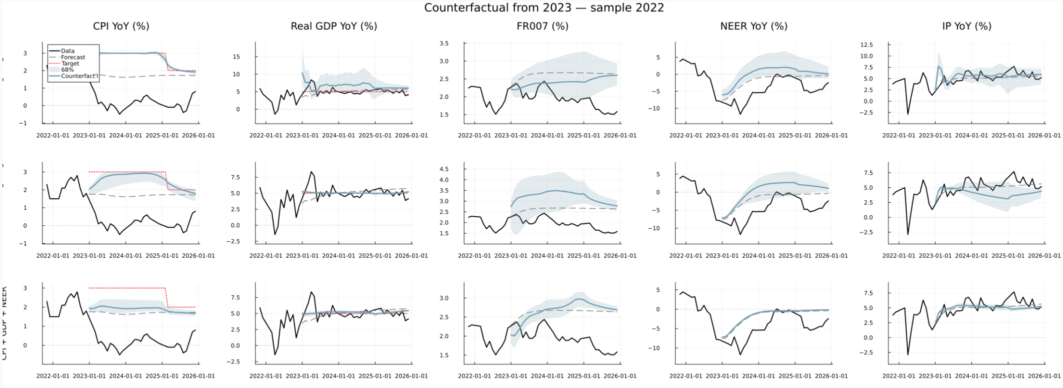
where

$$A_{stack} = \begin{bmatrix} \sqrt{\lambda_\pi} M_\pi \\ \sqrt{\lambda_y} M_y \\ \sqrt{\lambda_i} M_i \\ \sqrt{\lambda_e} M_e \end{bmatrix}, \qquad b_{stack} = \begin{bmatrix} \sqrt{\lambda_\pi}(\pi^* - \pi^{\text{base}}) \\ \sqrt{\lambda_y}(y^* - y^{\text{base}}) \\ \sqrt{\lambda_i}(i_t - i_{t-1}) \\ \sqrt{\lambda_e}(0 - e^{\text{base}}) \end{bmatrix}.$$

Algebraically, $\lambda \cdot I = W$ which is the weighting matrix used in the calculation of $\hat{A} = M \cdot W$ for the rules written in the form of $\hat{A}_x \tilde{x} + \hat{A}_z \tilde{z} = 0$ used in Caravello et al. 2025.

Results

Headline Results



Counterfactual paths using data until 2022

The first row corresponds to the strict CPI targeting with a smoothness penalty on interest rate changes: $\lambda_\pi = \lambda_i = 1$;
The second row adds the penalty for GDP deviation: $\lambda_\pi = \lambda_y = \lambda_i = 1$;
The third row adds the penalty for exchange rate volatility: $\lambda_\pi = \lambda_y = \lambda_i = \lambda_e = 1$.

▸ Counterfactual paths using data until 2025

Takeaways

- Across all three targeting rules, the price puzzle is uniformly present. In the first one with only CPI targeting, a persistent rate cuts compared to the forecast is required to hit the CPI target, but as a result the GDP would be higher than the baseline and the target. With both CPI and GDP targeting, a sustatining rate increased is necessary. As a result, however, the industrial output would suffer a sharp decline. With the addition of exchange rate stability rule, the rate path is similar to the forecast, alongside output and inflation. Still, the interest rate and exchange rate in all three rules is higher than the reality, which is not something that the government would be happy to see.
  - Unfortunately, as mentioned before, since net export, consumption and investment are used in the monthly GDP construction, we did not have it in the reduced form forecast and thus cannot calculate the counterfactual path to analyze the effects on them.
- The reduced-form baseline forecast from 2023 onward using data until 2022 actually did a good job in the predicting the GDP growth and the industrial output, while the CPI forecast is much higher than the reality, and this issue is only partially resolved by using the data until 2025, suggesting that the model is not able to capture the structural changes in the economy.

Limitations and Extensions

Limitations and Extensions

1. **Fragmented fiscal authority across regions**

   - Caravello et al. (2025) show that for the US, the empirics-only counterfactual is close to those produced by models like RANK and HANK.
   - This relies on a unified fiscal–monetary stance: the IRFs span the relevant policy frontier because there is one fiscal authority behind them. In China (provinces, LGFVs) and the euro area (member states, NGEU), the IRFs we estimate average over heterogeneous fiscal actors, and the spanning argument is no longer clean.

2. **Fiscal Regime Changes**

   - Caravello et al. assumed fiscal policy fixed along the counterfactual monetary path. In China this is hard to defend: the 2014 budget law reform, LGFV debt swaps, the 2019 VAT cuts, and the 2024 local-government debt resolution all shifted the fiscal stance materially within the sample.

Limitations and Extensions (cont'd)

3. **Where I would like to go next.**

- Both extensions point back to the question that motivated this thesis in the first place: what actually determines the price level when the monetary–fiscal split is non-standard?
    - **Monetary–fiscal interaction.** The central bank takes the fiscal stance as given, the fiscal authority takes the rate path as given, and the price level is determined jointly.
    - **Fiscal–fiscal interaction.** When "fiscal" is itself a collection of actors, central vs. provincial in China, federal-like vs. member-states in the euro area, their strategic interaction shapes the aggregate fiscal stance that the central bank then takes as given.
- The euro area is the natural next laboratory: institutionally between the unified US benchmark and the Chinese case, and with the data quality needed to credibly identify the joint monetary–fiscal object.

Appendix

Appendix — Asymmetric GDP Gap Specification

Construction. Define the threshold indicator $\mathbb{1}_t^- = \mathbb{1}\{\text{gdp\_gap}_t < 1\}$, then split:

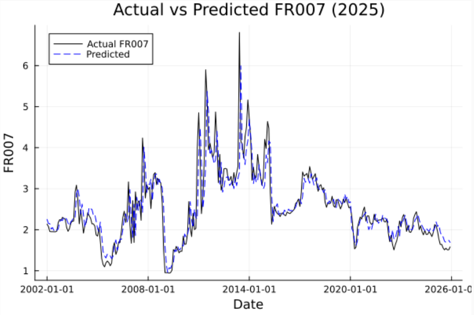
$$\text{gap\_pos}_t = \text{gdp\_gap}_t \cdot (1 - \mathbb{1}_t^-), \qquad \text{gap\_neg}_t = \text{gdp\_gap}_t \cdot \mathbb{1}_t^-.$$

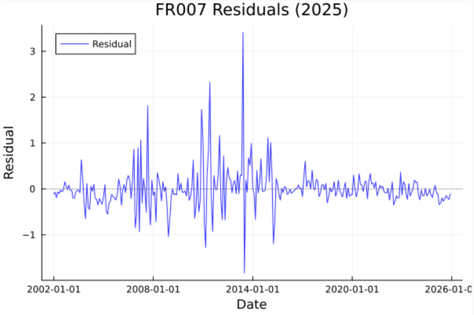
Motivation. Following Chen, Ren & Zha (2018), the PBoC's response to growth is hypothesised to be asymmetric. MANR2026 further estimated this threshold at around 1pp, meaning that the PBoC tends to tolerate small overshoots in output.

back

Appendix — Policy Rule Fit and Residuals

Left Actual FR007 vs. predicted FR007 from the rule regression. **Right.** Estimated residual series $\widehat{\varepsilon}_t^{\mathrm{MP}}$, used as the narrative monetary policy shock.

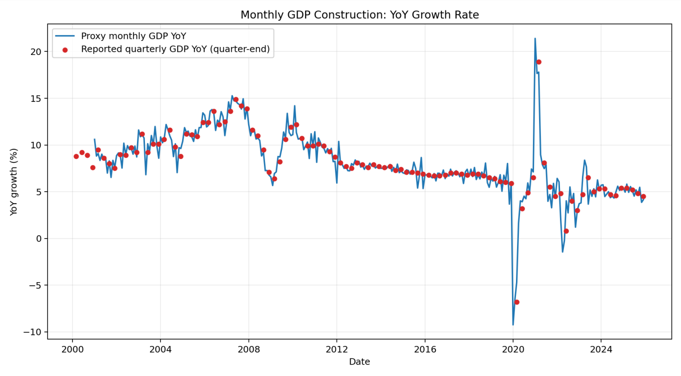




back

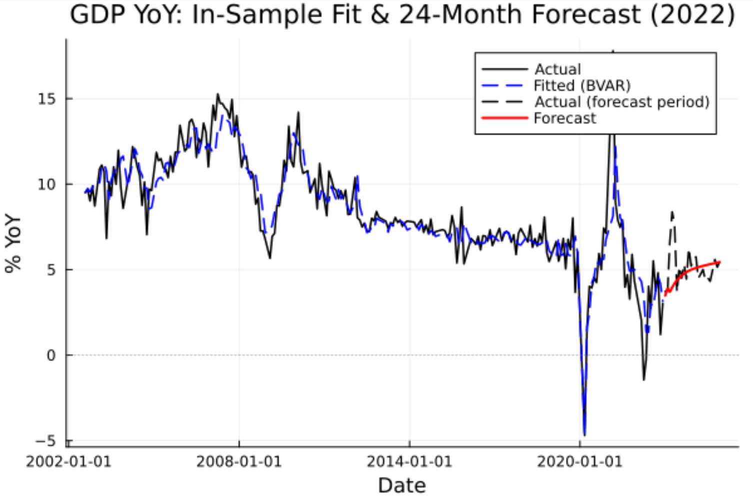
Appendix — Monthly GDP Construction

Stock & Watson (2010) distribution method, adapted to NBS quarterly GDP using the sum identity $Q_T = q_{3T-2} + q_{3T-1} + q_{3T}$, with monthly indicators Consumption / Fixed Assets Investment/ Government Spending / Trade Balance.



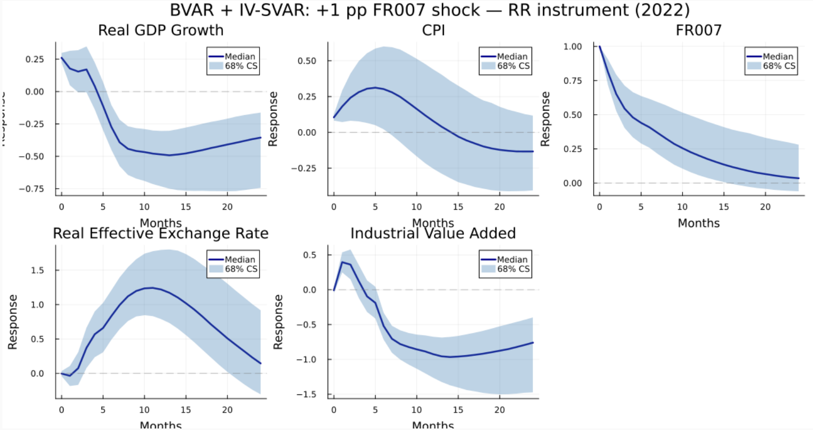
back

Appendix — Forecast vs. Actual (Proxy) Monthly GDP



back

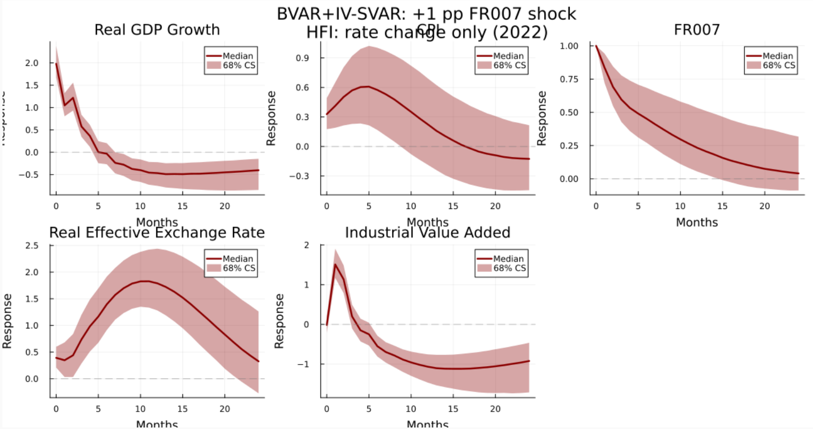
Appendix — Narrative Shock IRFs



Initial increases in CPI mirrors the results from MANR2026, which they attribute to the fact that Chinese CPI takes into account the mortgage payments as housing costs, and thus can rise in response to a monetary tightening very quickly.

back

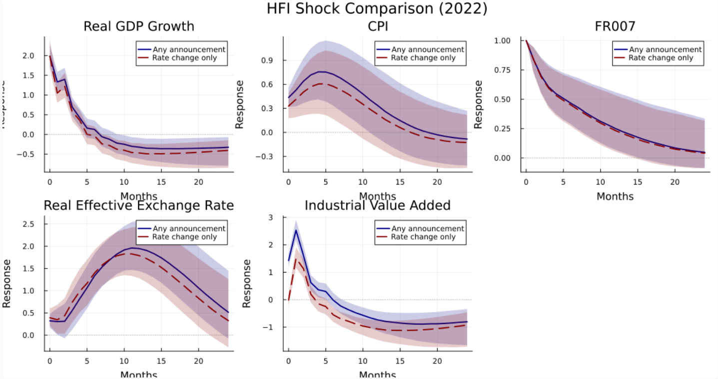
Appendix — HFI Shock IRFs



back

Appendix — HFI Shocks: IRF Comparison

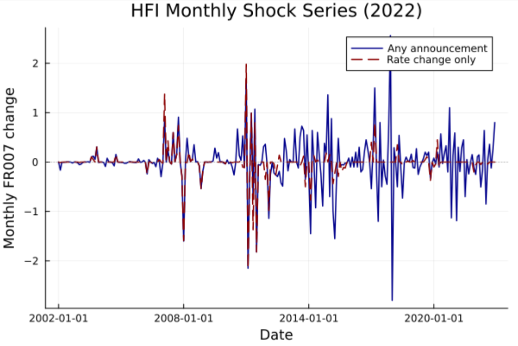
Blue. IRF of Chinese economy to monetary policy shocks identified from FR007 surprises on *any* PBoC policy announcement day. **Red.** IRF using *rate-change-only* dates.



back

Appendix — HFI Monthly Shock Series

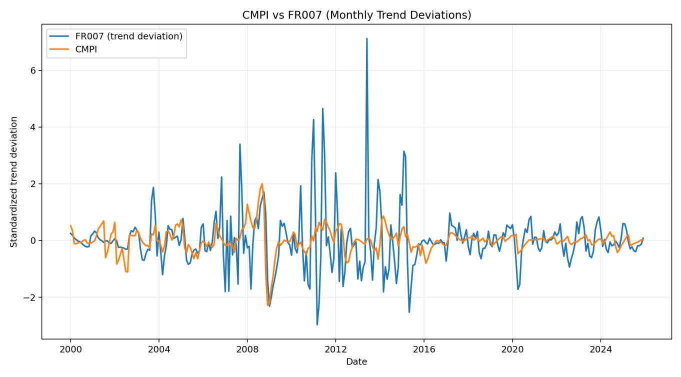
Monthly aggregation. Daily FR007 surprises are summed within each calendar month, yielding the two monthly HFI shock series used in the counterfactual analysis.



back

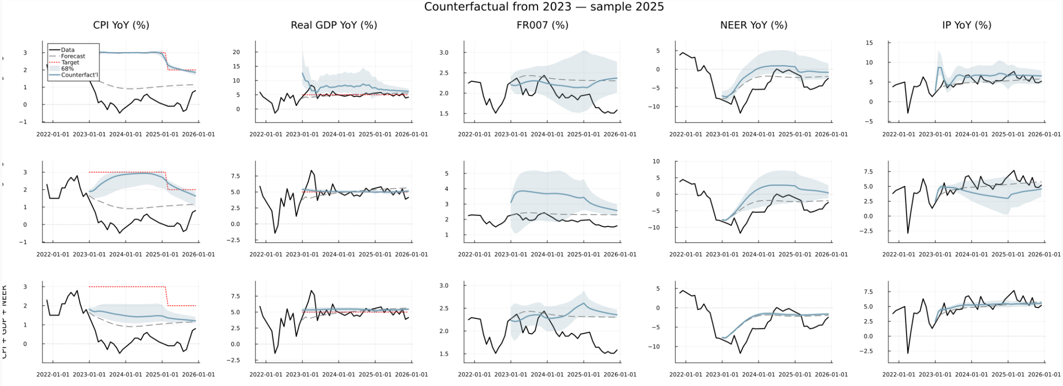
Appendix — CMPI vs. FR007

Specification choice. Comparison of policy indicators' deviations using FR007 versus CMPI (MANR2026).



The choice to retain FR007 can be also attributed to the fact that it contains enough variations, though less than CMPI, to identify the narrative shocks, and it is more directly linked to the HFI shocks identified from FR007 surprises.

Appendix —Headline Results with Data Until 2025



Counterfactual paths using data until 2025

back